

$P(h) \rightarrow$ probability of hypothesis h in the absence of any data

$P(D/h) \rightarrow$ probability that the given data would have generated if hypothesis h is true

$$P(h/D) = \frac{P(h)P(D/h)}{P(D)} \rightarrow \text{posterior probability}$$

$$\underline{P(\text{Play Tennis} = \text{Yes} / \text{Sunny})}$$

$$\underline{P(\text{Play Tennis} = \text{No} / \text{Sunny})}$$

if $P(\text{Yes} / \text{Sunny}) > P(\text{No} / \text{Sunny})$
then we say the player will
play tennis given sunny weather

What is a practical problem to using Bayesian learning? Example (classifying flowers)

X_1 = stalk length

X_2 = petal size

X_3 = petal colour

⋮

Y = type of flower

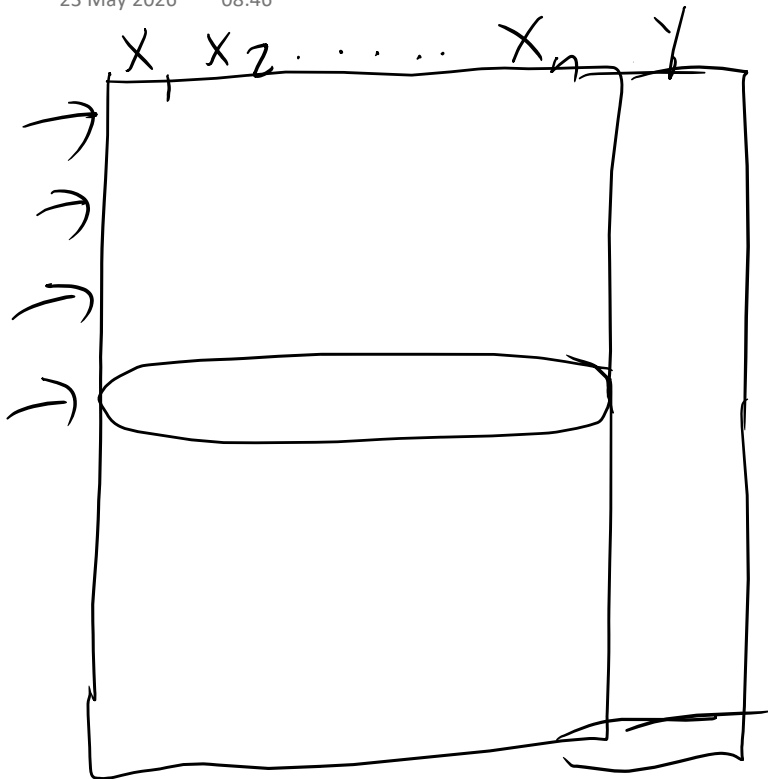
$$P(Y/X_1, X_2, \dots, X_n)$$

X_1	X_2	...	X_n	Y

$$P(Y/x_1, x_2, \dots, x_n) = \frac{P(Y, x_1, x_2, \dots, x_n)}{P(x_1, x_2, \dots, x_n)}$$

$$P(Y/x_1, x_2, \dots, x_n) = \frac{P(Y) P(x_1, x_2, \dots, x_n/Y)}{P(x_1, x_2, \dots, x_n)}$$

joint probabilities are
very hard to compute



$I(x_1 = \text{yes}, x_2 = \text{no}, x_3 = \text{yes} \dots)$

is decided by very few entries in this table

→ we may need to have an exponential entries before we see the particular instance we are interested in.

how do we simplify this?

$$P(X_1, X_2, Y) = P(Y) \cancel{P(X_1, X_2/Y)}$$

$$\cancel{P(X_1, X_2/Y)} = P(X_1/X_2, Y) P(X_2/Y)$$

$$\cancel{P(Y) P(X_1, X_2/Y)} = P(X_1/X_2, Y) \cancel{P(X_2/Y) P(Y)}$$

$$P(X_1, X_2, Y) = \frac{P(X_1/X_2, Y) P(X_2, Y)}{P(X_2, Y)}$$

$$P(X_1, X_2, Y) = P(Y) P(X_2/Y) P(X_1/X_2, Y)$$

$$P(A, B, C, D, \dots)$$

$$= P(A) P(B/A) P(C/A, B) P(D/A, B, C) \dots$$

Bayes Theorem factorization

Bayes theorem factorization

$$P(Y, x_1, x_2, \dots, x_n) \\ = P(Y)P(x_1/Y)P(x_2/x_1, Y)P(x_3/x_1, x_2, Y) \dots$$

We can write using conditional
independence

$$P(x_2/x_1, Y) = P(x_2/Y)$$

$$P(x_3/x_1, x_2, Y) = P(x_3/Y)$$

$$P(x_1, x_2, \dots, x_n/Y) = P(Y) \prod_{i=1}^n P(x_i/Y)$$

Naive Bayes Simplification

$$P(C_i/X) = \frac{P(X/C_i)P(C_i)}{P(X)}$$

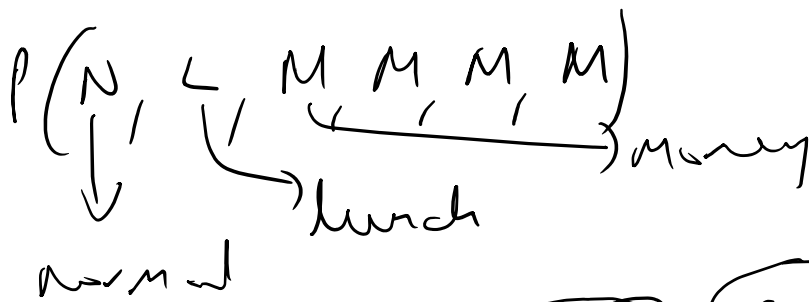
$$\sum_{i=1}^n P(C_i/X) = \sum_{i=1}^n \frac{P(X/C_i)P(C_i)}{P(X)}$$

$$\sum_{i=1}^n \frac{P(X \& C_i)}{P(X)}$$

$$\begin{aligned} P(X \cap (C_1 \cup C_2 \cup \dots \cup C_n)) &= P(X) \\ &= P((X \cap C_1) \cup (X \cap C_2) \dots \cup (X \cap C_n)) \\ &= P(X \cap C_1) + P(X \cap C_2) + \dots + P(X \cap C_n) \end{aligned}$$

because $X \cap C_i$ is disjoint from all other $X \cap C_j$

$$\begin{aligned} \therefore P(X \cap C_1) + P(X \cap C_2) + \dots + P(X \cap C_n) \\ = P(X) \end{aligned}$$



$$\begin{aligned}
 &= P(N) P(L/N) \underbrace{P(M/N, L)}_{\text{}} \underbrace{P(M/N, L, M)}_{\text{}} \underbrace{P(M/N, L, M, M)}_{\text{}} \\
 &= P(N) P(L/N) \underbrace{P(M/N) P(M/N) P(M/N) P(M/N)}_{[P(M/N)]^4}
 \end{aligned}$$

$\frac{N_{ic}}{N} \rightarrow$ original probability (could be zero)

$$\frac{N_{ic} + 1}{N + T}$$

add a fake document
containing word i

$0 < \frac{N_{ic} + 1}{N + T} < 1$ $\nexists$ also add a fake
document containing
each of the words in
the vocabulary.

we have
already that $N_{ic} < N$

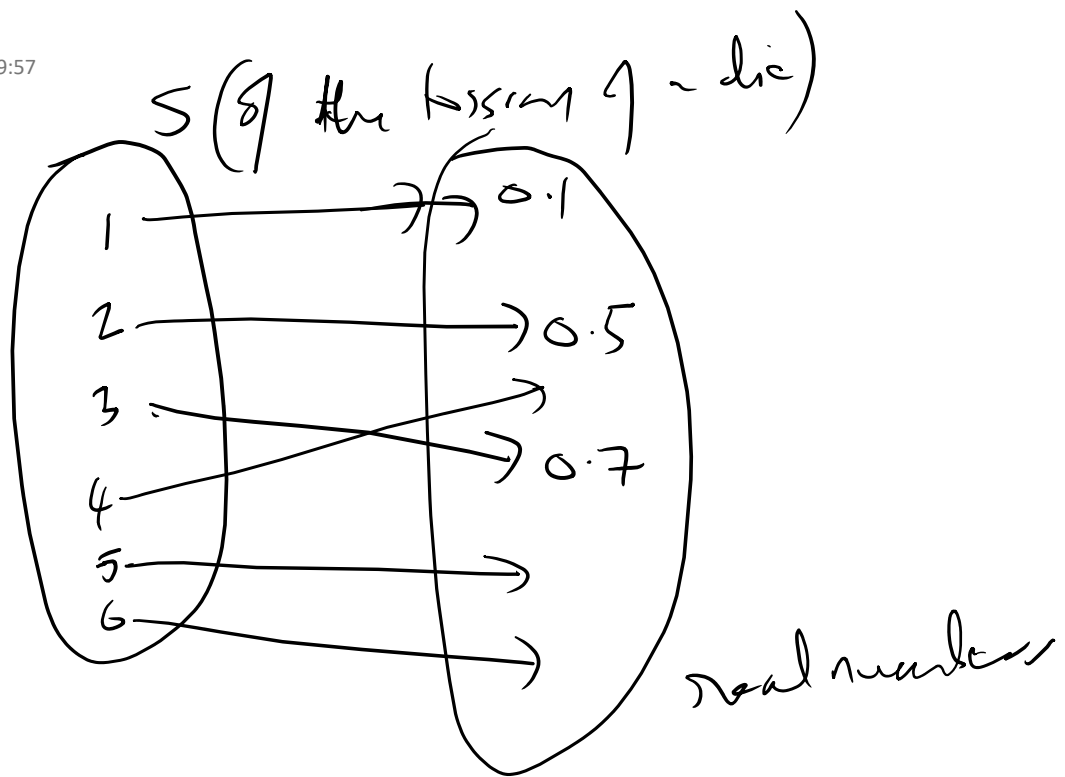
if we add some quantity to the
numerator, and a larger quantity
to the denominator we preserve
the property that the fraction < 1

Random variables

Two die $\rightarrow$ toss them

$(2, 1)$ $(5, 6)$ are possible outcomes
we are looking at the top faces of
the two die.

A random variable is a number
associated with an outcome of
an experiment



$x(\varepsilon_i)$ is a number, must be specified for every ε_i
 $\hookrightarrow \underline{x_{s_i}}$

Note that $\varepsilon_i = \text{outcome}$.

Let X be a random variable. We can talk of the probability that $X \leq a$

$$P\{X \leq a\}$$

$$\underline{P\{X \leq a\} ?}$$

$\{X \leq a\}$ is an event consisting of all outcomes ϵ_i such that $\underline{X(\epsilon_i) \leq a}$

$P\{X \leq a\} =$ sum over all the probabilities of all such outcomes ϵ_i

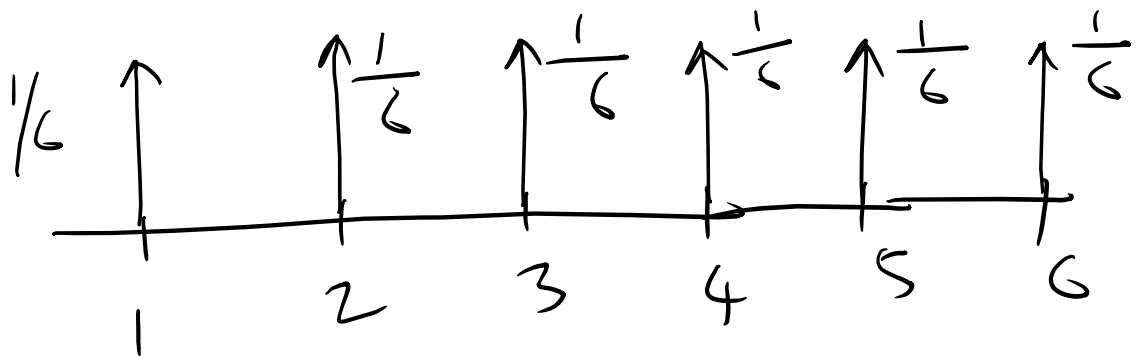
$$P\{X \leq a\} = \sum_{\substack{\epsilon_i \text{ such that} \\ X(\epsilon_i) \leq a}} P(\epsilon_i)$$



$$P\{X = 1\} = \frac{1}{6} \quad P(1)$$

$$P\{X = 2\} = \frac{1}{6} \quad P(2)$$

$$\underline{P\{X \leq 4\} = \{1, 2, 3, 4\} = \frac{4}{6}}$$



$$\sum \frac{1}{6} = 1$$

As the random variable X takes all possible values, we account for all outcomes in the sample space

$$\sum_{\text{over all } x_i} P\{X = x_i\} = \underline{P(\mathcal{S}) = 1}$$

$$P\{X = x_i\} = 1 \left(\frac{\text{branch 1}}{\text{outcomes}} \right)$$

$$\{X = x_i\} \cap \{X = x_j\} = \emptyset$$

$$\begin{aligned} P\{(X = x_1) \cup (X = x_2) \dots \cup (X = x_n)\} \\ = P\{X = x_1\} + P\{X = x_2\} + \dots + P\{X = x_n\} \\ = 1 \end{aligned}$$

Compute the average value of die tosses

Empirically look at values obtained from 100 tosses of the dice and take the average.

Theoretically we can do

$$\mu = \sum x f(x) = 1 \cdot \left(\frac{1}{6}\right) + 2 \cdot \left(\frac{1}{6}\right) + 3 \cdot \left(\frac{1}{6}\right) + 4 \cdot \left(\frac{1}{6}\right) + 5 \cdot \left(\frac{1}{6}\right) + 6 \cdot \left(\frac{1}{6}\right)$$

$$= \frac{1}{6} [1 + 2 + 3 + 4 + 5 + 6]$$

$$= \frac{1}{6} \times \frac{6 \times 7}{2} = \frac{7}{2} = \underline{3.5}$$

$$E(X) = \sum_{\substack{\text{over all} \\ \text{possible values of} \\ x}} x f(x)$$

$$E(\underbrace{aX}_y) = \sum_{\substack{\text{over all} \\ \text{possible values of} \\ x}} a x f(x)$$

In $E(aX)$ the expectation is over all possible values of X

$$E[g(x)] = \sum g(x_i) f(x_i)$$

↳ sum is over all possible values x can take

$$E(aX) = \sum a x_i f(x_i) = a \sum x_i f(x_i) = a E(X)$$

$$E(X + b) = \sum (x_i + b) f(x_i)$$

$$= \underbrace{\sum x_i f(x_i)}_{E(X)} + \underbrace{\sum b f(x_i)}_{b \underbrace{\sum f(x_i)}_{=1}}$$

$$= \underline{E(X) + b}$$